QP CODE 2590078

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-BCA (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

Core Course (CC) - MG1CCRBCA101 - DISCRETE MATHEMATICS

(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION

Part A

Short Answer Type Questions

Answer any 5 Questions Each Question carries 2 marks

1. Addition Rule - 2 marks
2. A bit string is a sequence of bits, where each bit is either 0 or 1. - 1 mark
   Give any bit string of length 9 - 1 mark
3. Correct Definition - 1 mark
   $|A \cap B| = |A|$ - 1 mark
4. Definition - 2 marks
5. Definition - 1 mark
   Example - 1 mark
6. $R_1 \cap R_2 = \{(1,1)\}$ - 1 mark
   $R_1 - R_2 = \{(2,2), (3,3)\}$ - 1 mark
7. $\begin{vmatrix} 5 & 3 \\ -1 & 2 \end{vmatrix} = 10 - -3 = 13 \neq 0$ - 1 mark
   non singular - 1 mark
8. $6A = \begin{bmatrix} 12 & 18 \\ 6 & 12 \end{bmatrix}$ - ½ mark
   $3B = \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix}$ - ½ mark
   $6A + 3B = \begin{bmatrix} 24 & 24 \\ 9 & 21 \end{bmatrix}$ - 1 mark

Part B

ShortEssayType Questions

Answer any **5** Questions

Each Question carries **6** marks

9.

p	q	p→q	q→p	(p→q)→(q→p)
T	T	T	T	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

For correct table give 6 marks

10. For each explanation give 2 marks

11. a) {1,2,3,4,5,6,7,8,9,10,11} - 2 marks
    b) null set or Φ - 2 marks
    c) {1,3,5,7,9} - 2 marks

12. Definition -1 mark
    Table construction - 2 marks
    Correct Graph - 3 marks

13. Not symmetric - 1 mark Explanation - 2 marks
    Not antisymmetric - 1 mark Explanation - 2 marks
    Write any one give 3 marks

14. **R** is reflexive with explanation (relation is reflexive if every person is related to themselves. So, a and a definitely speak a common language).
    - 1 ½ marks

    R is symmetric with explanation (If a and b speak a common language, then automatically b and a speak the same common language)
    - 1 ½ marks

    R is not transitive with explanation (Let a and b speak one language. Let b and c speak another language. It does not follow that a and c speak a Common language - 1 ½ marks
    Since R is reflexive and symmetric, but not transitive. Therefore, **R** is not an equivalence relation - 1½marks

15. Formation of matrices - ½ mark
    Finding IAI =8 - 1 mark $|A_1| = 8$ - 1 mark
    $|A_2| = 16$ - 1 mark $|A_3| = -8$ - 1 mark
    Solutions $x = |A_1|/|A| = 1$, $y = |A_2|/|A| = 2$, $z = |A_3|/|A| = -1$ -1½ marks

16. Formula $A^{-1} = \text{Adj } A/|A|$ - 1 mark
    $|A| = 1$ - 1 mark

$$\text{Cofactor matrix} = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$ - 2 marks

$$\text{Adj } A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$ - 1 mark

$$A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$ - 1 mark

Part C

Essay Type Questions
Answer any **3** Questions
Each Question carries **10** marks

17. a) Definition of Tautology - 3 marks

b)

p	q	⌐p	¬q	p→q	⌐p∧ (p→q)	(¬p ∧ (p → q)) → ¬q
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	F
F	F	T	T	T	T	T

Qn is wrong. It is not a Tautology. For a reasonable attempt give 5 marks.
For correct table give 7 marks.

18. Correct definition of conjunction with truth table - 3 ½ marks
Correct definition of disjunction with truth table - 3 ½ marks
Correct definition of exclusive or with truth table - 3 marks
Write only definition give 1 ½ marks for each definition.
Give 2 marks for each truth table.

19. AUBUC = {0,1,2,3,4,5,6,7,8,9,10} - 3 marks
A∩B∩C = {4,6} - 3 marks
For each Venn diagram give 2 marks .

20. Conditions /Definition for equivalence relation. - ½ mark
Reflexive with proof give 1 ½ mark
Symmetric with proof give 1 ½ mark
Transitive with proof give 1 ½ mark
R is an equivalence relation -1 mark
[0] = {...,-6,-3,0,3,6,...}
[1] = {...,-5,-2,1,4,7,...}
[2] = {...,-4,-1,2,5,8,...} For each equivalence classes give 1 mark
Prooving it is a partition of the set give 1 mark .

21. Finding $AB = \begin{bmatrix} 2 & 2 & -2 \\ 3 & 5 & -2 \\ 1 & -3 & -1 \end{bmatrix}$ -3 marks

$AB^T = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 5 & -3 \\ -2 & -2 & -1 \end{bmatrix}$ -1 mark

$$A^T = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix}$$ - 1 mark

$$B^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 3 & -2 \\ -1 & 4 & -3 \end{bmatrix}$$ - 1 mark

$$B^TA^T = \begin{bmatrix} 2 & 5 & -3 \\ 2 & 3 & 1 \\ -2 & -2 & -1 \end{bmatrix}$$ - 3 marks

Verification $AB^T = B^TA^T$ - 1 mark

22. Formation $AX = B$ - 1 mark

Formula $X = A^{-1}B$ - ½ mark

Finding $A^{-1} = \frac{1}{2}\begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$ - 2 marks

$$A^{-1}B = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$ - 2 marks

Writing solutions a=2, b=1 , c=0 - ½ mark

b) $A^2 = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$ - 1 mark

$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}$ - ½ mark

$7I = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$ - ½ mark

Substitution and simplification and answer give - 2 marks.

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